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SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Symbiosis International (Deemed University)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Assignment No. 09

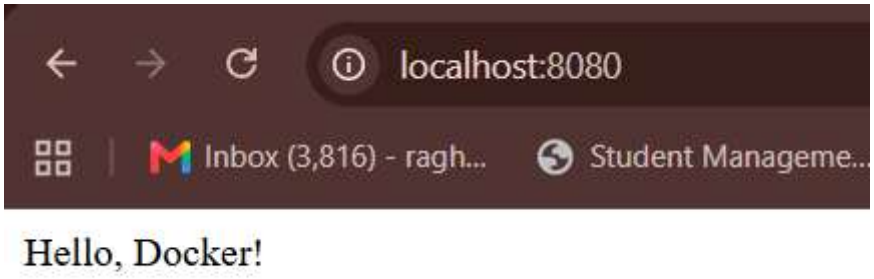
Subject:	DevOps Lab
TCODE:	TE7950
Name of Student	**Raghav Sharma**
PRN No.	**23070122171**
Branch	CSE, Batch (2023-27), L3
Academic Year & Semester	25 – 26, Sem VII
Date of Performance	23rd July, 2026
Title of Assignment:	Container Management & Networking
Theory:	Docker Container Management involves creating, running, stopping, inspecting, and removing containers throughout their lifecycle. Docker Networking enables communication between containers and external systems using port mapping and virtual networks. It ensures secure, isolated, and efficient communication while allowing applications running inside containers to be accessed from the host machine.

Implementation:

Step 1: Run the previously created Docker image in detached mode with port mapping:
docker run -d -p 8080:5000 my-flask-app

```
C:\Users\sharm\Desktop\7th sem\DevOpsL\myflaskproject>docker run -d -p 8080:5000 myflaskapp
c646289e3cbc5fa7ca6dfda7dfa7e0ef7ec25a0814cc76d859a694928b0d9cf7
```

Step 2: Open http://localhost:8080 and verify the application is running.



Step 3: Display all running containers using:
docker ps

```
C:\Users\sharm\Desktop\7th sem\DevOpsL\myflaskproject>docker ps
CONTAINER ID   IMAGE        COMMAND           CREATED          STATUS          PORTS                                       NAMES
c646289e3cbc   myflaskapp   "python app.py"   43 seconds ago   Up 42 seconds   0.0.0.0:8080->5000/tcp, [::]:8080->5000/tcp   zen_shtern
```

Step 4: Stop the running container using:
docker stop <container-id>

```
C:\Users\sharm\Desktop\7th sem\DevOpsL\myflaskproject>docker stop 57a53f58aebe
57a53f58aebe
```

Step 5: Display all containers using:
docker ps -a

```
C:\Users\sharm\Desktop\7th sem\DevOpsL\myflaskproject>docker ps -a
CONTAINER ID   IMAGE                     COMMAND                  CREATED              STATUS                          PORTS
          NAMES
c646289e3cbc   myflaskapp                "python app.py"          About a minute ago   Up About a minute               0.0.0.0:8
->5000/tcp   zen_shtern
f4ec9dd44cb0   myflaskapp                "python app.py"          About a minute ago   Created
          youthful_goodall
57aee8ead92d   myflaskapp                "python app.py"          3 minutes ago        Created
          quirky_maxwell
f8cdeed68200   myflaskapp                "python app.py"          7 minutes ago        Exited (0) 6 minutes ago
          romantic_mendeleev
```

Step 6: Inspect the container network configuration using:
docker inspect <container-id>

```
C:\Users\sharm\Desktop\7th sem\DevOpsL\myflaskproject>docker inspect c646289e3cbc
[
    {
        "Id": "c646289e3cbc5fa7ca6dfda7dfa7e0ef7ec25a0814cc76d859a694928b0d9cf7",
        "Created": "2026-07-23T04:25:07.954498281Z",
        "Path": "python",
        "Args": [
            "app.py"
        ],
        "State": {
            "Status": "running",
            "Running": true,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 6811,
            "ExitCode": 0,
            "Error": "",
            "StartedAt": "2026-07-23T04:25:08.0914250012",
            "FinishedAt": "0001-01-01T00:00:00Z"
```

Step 7: Remove the stopped container using:
docker rm <container-id>

```
C:\Users\sharm\Desktop\7th sem\DevOpsL\myflaskproject>docker rm 57aee8ead92d
57aee8ead92d
```


Subject Faculty Name: Prof. Arijit Dutta